

LETTER OPEN ACCESS

General Predictions for the Effects of Warming on Competition

Kaleigh E. Davis¹  | Tess N. Grainger¹  | Po-Ju Ke²  | Patrick L. Thompson³  | Mary I. O'Connor³  |
Joey R. Bernhardt¹ 

¹Department of Integrative Biology, University of Guelph, Guelph, Canada | ²Institute of Ecology and Evolutionary Biology, National Taiwan University, Taipei, Taiwan | ³Department of Zoology and Biodiversity Research Centre, University of British Columbia, Vancouver, Canada

Correspondence: Kaleigh E. Davis (kaleighedavis@gmail.com)

Received: 6 October 2025 | **Revised:** 17 April 2026 | **Accepted:** 21 April 2026

Editor: David Mouillot

Keywords: coexistence | coexistence theory | competition | global change | metabolic theory | warming

ABSTRACT

Understanding the effects of climate change on ecological communities has been limited by a lack of general theory for how temperature affects competition. To fill this knowledge gap, we integrated Modern Coexistence Theory and the Metabolic Theory of Ecology by incorporating empirically derived temperature sensitivities into Modern Coexistence Theory's central model. We then simulated warming in consumer-resource systems and found that warming reduced both niche and fitness differences, making species more ecologically similar and competitive interactions more neutral. The greatest shifts in competition occurred when temperature sensitivities among species were highly asymmetrical. Effects of warming on competition via niche differences were comparable to those on fitness differences, suggesting that the emphasis on vital rates in global change research may overlook key biodiversity drivers. This general theory expands the domains of two prominent ecological theories and provides predictions for how warming may alter competition even in benign regions of species' thermal niches.

1 | Introduction

Understanding the effects of climate warming on biodiversity is a central challenge of modern ecology (Parmesan and Yohe 2003; Peters and Lovejoy 1992; Román-Palacios and Wiens 2020). Species interactions, and competition in particular, influence species' responses to the environment, and are thus critical to understanding the effects of warming on biodiversity (Godoy 2019; Tilman 1994). While a considerable body of empirical work has demonstrated shifts in competition with changing environments (e.g., Alexander et al. 2015; Germain et al. 2018; Pérez-Ramos et al. 2019), the nature of these shifts has been variable, and explanations usually context specific. Whether there are any general patterns in how warming affects competition, or whether these changes are inherently species specific, is a major open question. In the face of rapid environmental change, new theory is needed to address this question.

Modern Coexistence Theory has become the leading framework for understanding competition and coexistence (Chesson 2000; Godoy and Levine 2014). This theory has the advantage of being agnostic to the specific mechanisms or context of competitive interactions, which allows it to be used to investigate general principles of competition and to be applied across a wide range of systems (Godwin et al. 2020; Grainger et al. 2019). Within Modern Coexistence Theory, in the absence of environmental fluctuations, species' competitive differences are described along two axes: fitness differences and niche differences (Chesson 2000). Fitness differences are interspecific differences that give one species a competitive advantage and stem from differences between competing species in intrinsic growth rate and sensitivity to competition. Niche differences are differences that reduce the strength of interspecific competition relative to intraspecific competition, for example, differences in resource use or susceptibility to predators (Chesson 2000; Chesson and Kuang 2008).

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2026 The Author(s). *Ecology Letters* published by John Wiley & Sons Ltd.

The balance of these differences determines whether two species can coexist, one species will exclude the other, or they are competitively neutral (Adler et al. 2007). In global change research, work to understand the effects of warming on species abundance and distribution has predominantly emphasized shifts in vital rates, which relate to fitness differences (e.g., Merow et al. 2014; Ehrlén and Morris 2015; but see Adler et al. 2012; Usinowicz and Levine 2018), while the potential effects of warming on niche differences remain underexplored.

One criticism of Modern Coexistence Theory is that, because it is phenomenological, it is not well-suited to generate a priori predictions about how a given factor will affect competition and coexistence (Terry 2025). Indeed, the parameters used to estimate niche and fitness differences in classical Modern Coexistence Theory models describe the outcomes of competitive interactions but are not related explicitly to underlying mechanisms (Chesson 2000). However, niche and fitness differences can be alternatively calculated from their constituent mechanistic processes (Chesson and Kuang 2008; Letten et al. 2017), for example, parameters related to resource competition (Song et al. 2019). Fortunately, the effects of temperature on many of these mechanistic processes have also been investigated empirically (Dell et al. 2011; Rall et al. 2010). To evaluate how these temperature effects shift competitive dynamics with warming, the logical next step is to integrate empirically grounded temperature dependencies into Modern Coexistence Theory models.

The Metabolic Theory of Ecology is a widely used framework for understanding the effects of temperature on ecological processes (Brown et al. 2004). A central premise of this framework is that ecological processes, such as population growth and species interactions, all stem from metabolic processes, which control the uptake and transformation of resources into maintenance, growth and reproduction (Brown et al. 2004). In this way, metabolism fundamentally and predictably constrains higher order processes across systems, leading to a predictable ‘scaling’ of temperature effects across levels of organization. From this principle of metabolic scaling, the Metabolic Theory of Ecology has been used to predict the effects of temperature on population growth rates (Savage et al. 2004), population carrying capacities (Bernhardt, Sunday, and O'Connor 2018), consumer-resource interactions (O'Connor 2009; O'Connor et al. 2011), food webs (Gibert et al. 2022; Gilbert et al. 2014; Vasseur and McCann 2005) and ecosystem productivity (Enquist et al. 2003; Lopez-Urrutia et al. 2006; Welter et al. 2015). Despite this broad success, extending the metabolic scaling approach to competitive interactions has proven challenging, due to a conflict in focus between competition theory and the Metabolic Theory of Ecology.

While competition theory has focused primarily on changes in ecological outcomes driven by interspecific variation in common underlying traits (here, intra-process asymmetry, Figure 1) (e.g., Kraft et al. 2008; Weiher et al. 1998), the Metabolic Theory of Ecology has largely ignored interspecific variation in thermal traits, and focused instead on average differences in thermal responses across metabolic processes (here, inter-process asymmetry, Figure 1, but see Clegg and Pawar 2024; Gibert et al. 2022). Since our focal traits are thermal traits, we will refer to these intra- and inter-process thermal asymmetries (Figure 1). Progress

to incorporate interspecific variation in thermal traits into metabolic scaling studies of competition has been made primarily by exploring consequences of high, above optimal, temperatures and generalist-specialist trade-offs in thermal tolerance breadth (e.g., Amarasekare and Johnson 2017; Simon and Amarasekare 2024; Smith and Amarasekare 2018). However, most species live, on average, well below their thermal optima in nature (Bernhardt, Sunday, Thompson, and O'Connor 2018; Martin and Huey 2008), and whether interspecific trait differences in the non-stressful (sub-optimal) temperature range alone can produce general patterns in competitive dynamics under warming is an open question. Incorporating temperature-dependence at sub-optimal temperatures into the governing processes of Modern Coexistence Theory enables us to answer this question.

Here we draw from the Metabolic Theory of Ecology and Modern Coexistence Theory to develop theory for how warming affects competition for shared resources at suboptimal temperatures. We use this theory to address three questions: (1) How do the temperature sensitivities of each process underlying competition contribute to changes in niche and fitness differences with warming? (2) How do inter- and intra-process thermal asymmetries affect these changes? (3) Are there general effects of warming on competition that emerge from the combined effects of warming on competition's suite of underlying mechanistic processes? To answer these questions, we analysed a MacArthur consumer-resource model (MacArthur 1970) using the canonical two-species Lotka-Volterra formulation of Modern Coexistence Theory that expresses phenomenological competition coefficients with mechanistic parameters (Chesson 1990, 2011; Chesson and Kuang 2008; Song et al. 2019). We modelled the effects of warming on competition by making each parameter in the MacArthur consumer-resource model temperature-sensitive. We then synthesized temperature sensitivity data for each model parameter from the published literature to generate an empirical distribution of temperature sensitivities, and used this distribution to simulate effects of warming on competition between species pairs. In doing so, we identified a novel set of general predictions for how competition is affected by warming, and the conditions under which we expect to see more extreme shifts in competition.

2 | Materials and Methods

2.1 | Resource Competition Model

We modelled competition in a two-consumer system (N_1 and N_2) with two substitutable resources (R_a and R_b) following the classic MacArthur consumer-resource model (MacArthur 1970):

$$\frac{dN_i}{dt} = \sum_{k \in \{a,b\}} v_{ik} c_{ik} R_k N_i - m_i N_i \quad (1)$$

$$\frac{dR_k}{dt} = r_k R_k \left(1 - \frac{R_k}{K_k}\right) - \sum_{i=1}^2 c_{ik} R_k N_i, \quad (2)$$

with $i=1$ or 2 and $k=a$ or b (see Figure S1 for simulations with >2 resources). Here, the resource follows logistic growth with intrinsic growth rate r_k and carrying capacity K_k . Consumer species i uses resource k following a Holling type-I functional response,

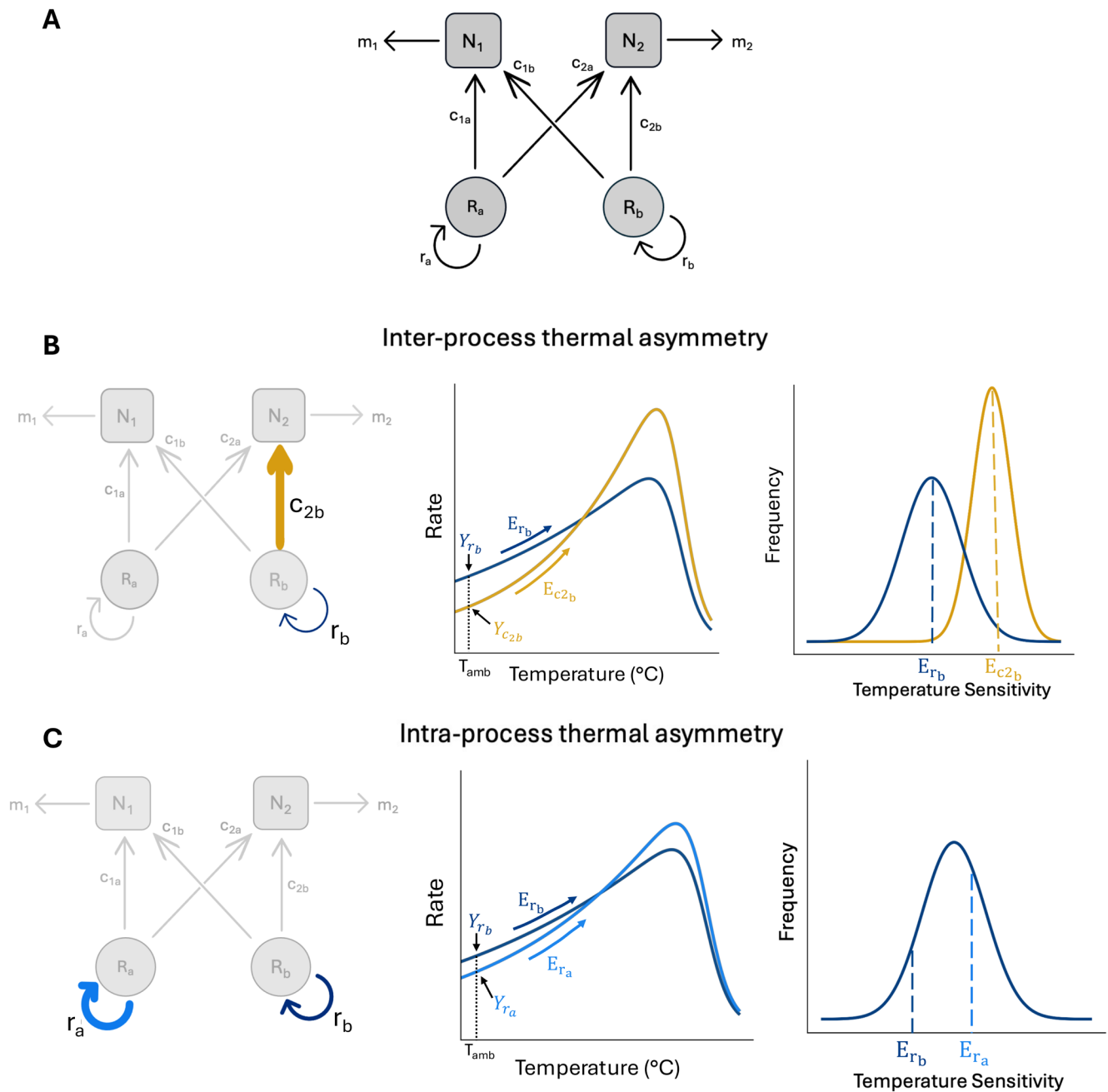


FIGURE 1 | Schematic of a consumer-resource model with temperature sensitive processes, yielding inter- and intra-process thermal asymmetries. In panel A, two unique consumers are shown as boxes (N_1 and N_2) and two substitutable resources as circles (R_a and R_b). Arrows show directional energy flow processes and are labelled with the model parameters they represent: m_i is the mortality rate of consumer i , c_{ik} is consumption rate of resource k by consumer i , and r_k is the intrinsic growth rate of resource k . Parameters K_k , resource carrying capacity, and v_{ik} , conversion efficiency, are not shown, but mediate the abundance of available resource, R_k , and the transfer of energy from R_k to N_i , respectively. Panels B and C show how temperature sensitivity, E , or the change in the rate of a process with temperature below the temperature where the curve peaks (central panels), is incorporated into the model via biological rates of the consumer and resource, and how this can generate two types of thermal asymmetries depending on where in the consumer-resource system a difference in temperature sensitivity occurs (left panels). The ambient temperature is given as T_{amb} , and the rate of each process at this temperature, corresponding to $Y_{T_{amb}}$ in Equation (9), is given. The thickness of the arrows in the left panels corresponds to the temperature sensitivity depicted in the central panels. The right panels in B and C relate a single observation of temperature sensitivity (dashed line) for a parameter to the broader empirical distribution of temperature sensitivities for that parameter, constructed from our data synthesis. In an inter-process thermal asymmetry (Panel B), the two temperature sensitivities come from independent distributions of temperature sensitivity in each parameter, whereas in an intra-process thermal asymmetry (Panel C), the two temperature sensitivities come from the same distribution of temperature sensitivity.

with per capita consumption rate (hereafter “consumption rate”) parameter c_{ik} and conversion efficiency parameter v_{ik} . While the type-I functional response is sometimes considered unrealistic,

recent work suggests that it may be appropriate in more cases than previously thought (Coblentz et al. 2023; Novak et al. 2025). Finally, consumers experience per-capita mortality m_i . We

converted this into a two-species Lotka–Volterra competition model by separation of timescales. Specifically, by assuming fast resource dynamics, the quasi-equilibrium of R_k can be solved as:

$$\widehat{R}_k = \frac{K_k}{r_k} \left(r_k - \sum_{i=1}^2 c_{ik} N_i \right), \quad (3)$$

which is a function of consumer abundances. Substituting this expression back into Equation (1), we arrive at the following:

$$\frac{dN_i}{dt} = N_i \left[\left(\sum_{k \in \{a,b\}} v_{ik} c_{ik} K_k - m_i \right) - \left(\sum_{k \in \{a,b\}} v_{ik} \frac{K_k}{r_k} c_{ik}^2 \right) N_i - \left(\sum_{k \in \{a,b\}} v_{ik} \frac{K_k}{r_k} c_{ik} c_{jk} \right) N_j \right] = g_i N_i \left[1 - \sum_{j=1}^2 \alpha_{ij} N_j \right] \quad (4)$$

with

$$g_i = \sum_{k \in \{a,b\}} v_{ik} c_{ik} K_k - m_i \quad (5)$$

$$\alpha_{ij} = \frac{\sum_{k \in \{a,b\}} v_{ik} \frac{K_k}{r_k} c_{ik} c_{jk}}{g_i} = \frac{\sum_{k \in \{a,b\}} v_{ik} \frac{K_k}{r_k} c_{ik} c_{jk}}{\sum_{k \in \{a,b\}} v_{ik} c_{ik} K_k - m_i}. \quad (6)$$

Here, g_i represents consumer i 's intrinsic growth rate, which equals the maximum energy gain by consumption minus the consumer's per-capita mortality rate. On the other hand, α_{ij} represents the relative reduction in consumer species i 's intrinsic growth rate caused by competition from species j (i.e., relative competition coefficient; *sensu* Song et al. 2019).

Modern Coexistence Theory then defines niche overlap (ρ) and fitness ratio ($\frac{f_2}{f_1}$) as follows:

$$\rho = \sqrt{\frac{\alpha_{12} \alpha_{21}}{\alpha_{11} \alpha_{22}}} \quad (7)$$

$$\frac{f_2}{f_1} = \sqrt{\frac{\alpha_{11} \alpha_{12}}{\alpha_{22} \alpha_{21}}}. \quad (8)$$

Here, ρ quantifies the relative strength of intraspecific and interspecific competition, and $\frac{f_2}{f_1}$ quantifies species' relative sensitivity to competition. Finally, we define niche differences (ND) as $-\log(\rho)$ and fitness differences (FD) as $\log\left(\frac{f_2}{f_1}\right)$. While other definitions of ND and FD are more flexible (Spaak et al. 2023; Spaak and De Laender 2020), the definitions we use here allow us to compare ND and FD on the same scale and to recover the ecologically intuitive condition for species coexistence, that is, the ND between a pair of species must be greater than their FD for them to coexist. These definitions have also been used extensively elsewhere in the literature (e.g., Johnson et al. 2022; Magee et al. 2025; Wan et al. 2025; Yamamichi et al. 2022).

2.2 | Adding Temperature Sensitivity to Model Parameters

For each mechanistic process from the MacArthur model (i.e., r_k , K_k , c_{ik} , v_{ik} , m_i ; Equation 4; Figure 1A), we modelled temperature sensitivity using an exponential Arrhenius-form function:

$$Y(T) = Y_{T_{amb}} e^{\left(\frac{E_Y}{k_B T_{amb}} - \frac{E_Y}{k_B T} \right)}, \quad (9)$$

where Y is any parameter of the consumer-resource model (Equation 4), T is the environmental temperature in Kelvin (K), T_{amb} is the ambient temperature, here set to 10°C (= 283.15 K), $Y_{T_{amb}}$ is the rate of the process at the ambient temperature, E_Y is the exponential rate of increase in process rate with increasing temperature (Figure 1; hereafter temperature sensitivity), and k_B is Boltzmann's constant (eV/K). The temperature sensitivity parameter, E_Y , is equivalent to the activation energy parameter from chemistry's Boltzmann-Arrhenius model, which has been applied to ecological rates within the Metabolic Theory of Ecology (Brown et al. 2004).

By standardizing environmental temperature, T , relative to ambient temperature, T_{amb} , we isolate the effects of 15°C of warming, rather than the particular range of temperatures employed for our analyses (10°C–25°C). While it is known that most individual- and population-level temperature responses are unimodal (Amarasekare and Savage 2012; Kingsolver 2009) and that exponential models can under- or over-predict the temperature sensitivity of biological processes (Knies and Kingsolver 2010; Michaletz and Garen 2024; Pawar et al. 2016), many studies have focused on the rate of increase in the exponential portion of the temperature-response curve (Figure 1B,C) to model scaling effects of temperature (Bernhardt, Sunday, and O'Connor 2018; Gibert et al. 2022; Gilbert et al. 2014). As such, we used an exponential model to describe warming-induced shifts in competition when both species were in this exponentially increasing portion of their thermal niche, and where the effects of temperature were non-stressful (sub-optimal). We sampled temperature sensitivity estimates for each parameter from posterior distributions (Figure 2) generated from an empirical data synthesis of the temperature sensitivity of each parameter (see Appendix S1).

2.3 | Simulating Effects of Warming on Competition

2.3.1 | Model Starting Conditions

We answered our three research questions by simulating competition across a 15°C temperature gradient (10°C–25°C) between consumer species pairs that coexist under ambient temperatures ($T_{amb} = 10^\circ\text{C}$). All simulations presented in the main text had the same starting conditions at T_{amb} , captured by the intercepts in the Arrhenius models ($Y_{T_{amb}}$; Equation 9, Table S1) that comprise the temperature-dependent consumer-resource model (Equation 4): at T_{amb} , consumer species were predicted to coexist and had symmetrical resource preferences, so that each consumer specialized to the same degree on one of two substitutable resources (given by c_{ik} and v_{ik} ; Table S1). At T_{amb} the preferred resource of N_2 (here, r_a) had a faster growth rate than the preferred resource of N_1 (here, r_b). This set of starting conditions placed our species pairs on the boundary between coexistence and competitive exclusion

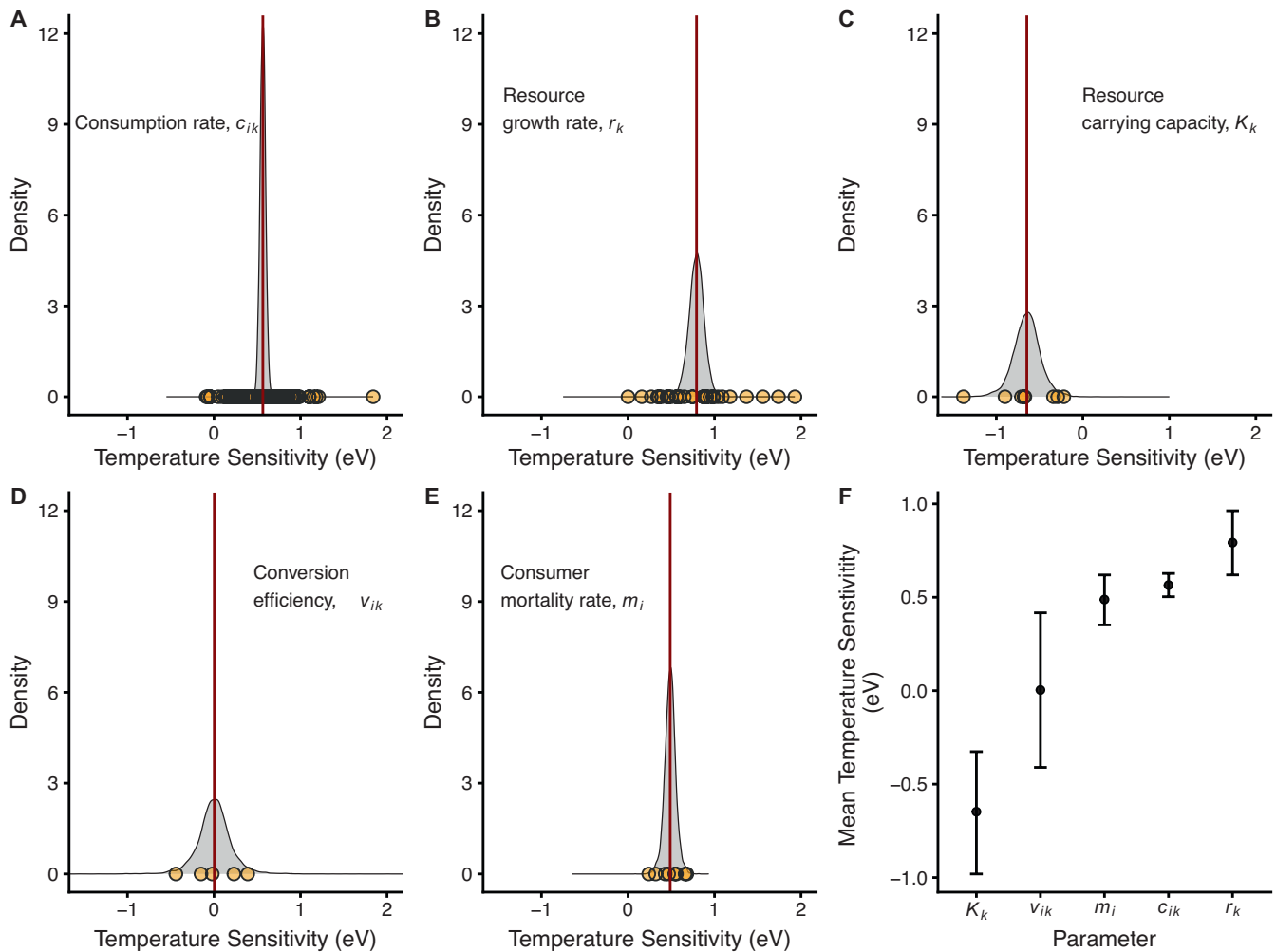


FIGURE 2 | Posterior distributions for temperature sensitivity of consumer-resource model parameters. Panels A–E show published empirical observations for each parameter in the consumer resource model (Equation 4) in orange circles and posterior distributions informed by these observations in grey density plots. Red lines indicate the mean of the posterior distribution. Panel F shows the means and 95% credible intervals (CIs) for each parameter.

of N_1 by N_2 (Figure 3). Other starting conditions for competing consumers that coexist under ambient temperature conditions were explored in supplementary analyses (Table S1, Figures S2–S4). Although 15°C of warming is unrealistic for changes in average temperature from a climate change perspective, we used this large thermal gradient to demonstrate our results clearly. Applying a more realistic thermal gradient (5°C) generated qualitatively similar results, but with less contrast (Figure S5).

2.3.2 | Simulation: Questions 1 and 2

To examine the effects of each parameter's temperature sensitivity on ND and FD (Question 1) and the effects of thermal asymmetries on competition (Question 2), we ran 500 warming simulations, varying one focal parameter at a time. For each simulation we randomly drew temperature sensitivity, E_y , values for the focal parameter from its empirical posterior distribution, and we set all other parameters' temperature sensitivities equal to 0. After each temperature sensitivity for the

focal parameter was drawn, we simulated 15°C warming and calculated ND and FD every 0.1°C, according to Equations (7) and (8). For resource growth rate, r_k , resource carrying capacity, K_k , and consumer mortality rate, m_i , temperature sensitivities for each consumer or resource were drawn independently for each simulation. For consumption rate, c_{ik} , we drew temperature sensitivities independently for each species' consumption rate on each resource (See Figure S6 for other draw configurations). Finally, for resource conversion efficiency, v_{ik} , we assumed that the temperature sensitivity primarily reflects the nutritional composition of the resource and therefore was the same for both consumers of a given resource (i.e., $E_{v_{1a}} = E_{v_{2a}}$).

To explore the effects of inter- and intra-process thermal asymmetries on competition (Question 2), we quantified the magnitude of the thermal asymmetries from the temperature sensitivities drawn for each simulation and examined the associated shift in ND–FD space (See Appendix S1). We quantified the magnitude of a thermal asymmetry as the absolute difference between the values drawn for the focal parameter in

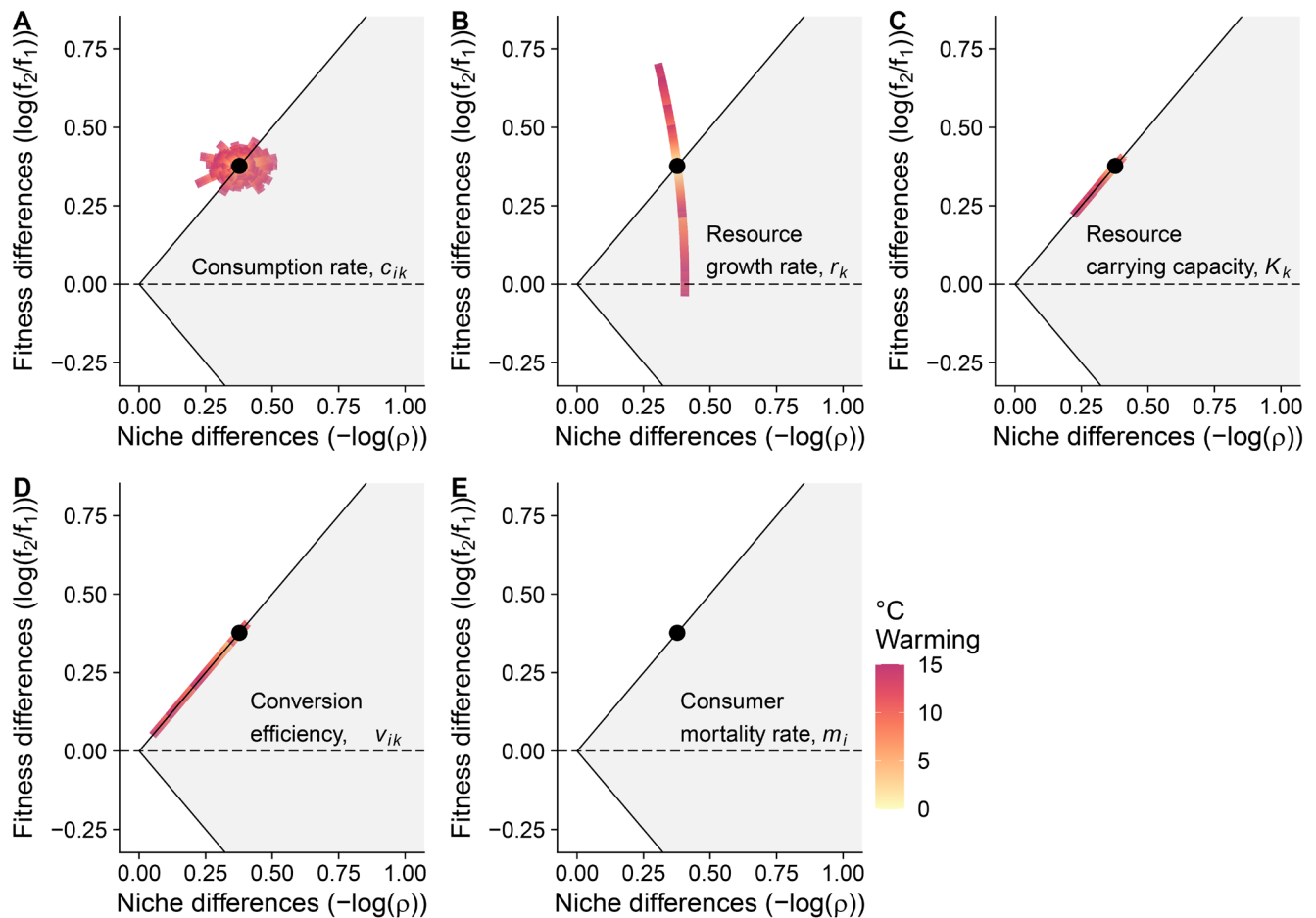


FIGURE 3 | Effects of temperature sensitivity in each model parameter on competition with warming. In each panel, temperature sensitivities of the focal parameter are drawn randomly from the parameter's empirical distribution (Figure 2) and then the system is warmed 15°C, while temperature sensitivities for all other parameters are held at 0. This is repeated 500 times and each simulated warming path is plotted. All simulations begin at the same ambient temperature, 10°C (black point). For all parameters except for consumption rate (i.e., panels B–E), the number of temperature sensitivities drawn for each simulation is two—one for each consumer or resource species (Figure 1A). This results in perfectly overlaid trajectories for each simulation: Note different lengths of temperature gradient paths, which end with a dark red colour. For consumption rate, in panel A, there are four temperature sensitivities drawn for each simulation, corresponding to one temperature sensitivity per resource, for each species (Figure 1A). As a result, warming trajectories do not fall perfectly on top of one another. The shaded region in the plots shows where species can coexist, that is, where $ND > |FD|$, and the unshaded regions show where one species competitively excludes the other, that is, $ND < |FD|$.

each simulation (e.g., $|E_{r_a} - E_{r_b}|$). We then quantified the shift in ND–FD space as the Euclidean distance between the model's starting position and the position after 15°C simulated warming.

2.3.3 | Simulation: Question 3

To identify general effects of warming on competition (Question 3), we used the same set of starting conditions as in Questions 1 and 2 (Table S1), but imposed temperature sensitivity on all model parameters simultaneously, then simulated warming over a 15°C thermal gradient. For each simulation ($N=500$), we drew temperature sensitivity values, E_Y , for all parameters from their respective distributions, then calculated ND and FD at each 0.1°C step. We quantified effects of warming on competition based on the distance between the species' starting ND and FD values (at T_{amb} , 10°C) and their median ND and FD values at the final temperature (25°C).

Data and code for analyses are available at DOI: [10.5281/zenodo.19634310](https://doi.org/10.5281/zenodo.19634310).

3 | Results

3.1 | Empirically-Derived Temperature Sensitivity of Parameters

We synthesized 157 temperature sensitivity estimates for consumer-resource parameters from 58 studies, representing 79 ectothermic consumer or resource species. Represented species included heterotrophic and autotrophic organisms as well as unicellular and multicellular organisms (Table S2, Figure S7). We found considerable variation within and across temperature sensitivity estimates, E_Y (Figure 2; Table S3), suggesting both inter-process and intra-process thermal asymmetries were present in our synthesized dataset. Posterior distributions and 95% credible

intervals for the temperature sensitivities of resource growth rate, r_k , consumption rate, c_{ik} , and consumer mortality rate, m_i , were positive, indicating that these rates increase with warming. Posterior means and 95% credible intervals were: $E_{r_k} = 0.79 \text{ eV}$ (0.62, 0.96 eV; 30 observations); $E_{c_{ik}} = 0.56 \text{ eV}$ (0.50, 0.63 eV; 105 observations); $E_{m_i} = 0.49 \text{ eV}$ (0.35, 0.62 eV; 8 observations; Figure 2A,B,E). Overlapping 95% credible intervals across these three parameters suggest no inter-process thermal asymmetry among them (Figure 2F). Contrastingly, resource carrying capacity, K_k , had a negative temperature sensitivity, $E_{K_k} = -0.65 \text{ eV}$ (-0.98, -0.33 eV; 8 observations; Figure 2C), indicating that carrying capacities decrease with warming. This negative temperature sensitivity was statistically distinct from that of consumption rate, resource growth rate and mortality rate (Figure 2F). Finally, the posterior distribution for the temperature sensitivity of conversion efficiency had a 95% credible interval that spanned both negative and positive values and was centered near 0: $E_{v_{ik}} = 0.0036 \text{ eV}$ (-0.41, 0.42 eV; 5 observations; Figure 2D). Credible intervals suggested inter-process thermal asymmetries between conversion efficiency and consumption rate and resource growth rate, but not between conversion efficiency and resource carrying capacity or mortality rate (Figure 2F).

3.2 | Question 1: How Do Temperature Sensitive Processes Underlying Competition Affect Niche and Fitness Differences?

For competing consumer species that co-exist under ambient temperature conditions, warming the system caused every process except mortality to have large effects on niche and/or fitness differences (ND, FD; Figure 3). Effects of warming on resource consumption rate, c_{ik} , drove variation in both ND and FD (Figure 3A), whereas effects of warming on resource growth rate, r_k , primarily drove variation in FD (Figure 3B). We found that temperature sensitivity in these two processes had the largest potential to affect competitive outcomes in our model, because their temperature sensitivity can move species pairs along a trajectory that is far from parallel (and in some cases close to perpendicular) to the coexistence boundary as the system warms. In contrast, effects of warming on resource carrying capacity, K_k , and conversion efficiency, v_{ik} , drove variation parallel to the coexistence boundary (Figure 3C,D, Figure S3). In our focal analysis, where species pairs underwent warming from a starting position on the coexistence boundary, temperature-dependent carrying capacity and conversion efficiency drove the species toward increased functional similarity (i.e., toward the neutral point).

In supplemental analysis of coexisting species pairs with different resource preferences at ambient temperature, and therefore different starting points in ND—FD space (Table S1), effects of temperature-sensitivity in resource consumption rate, c_{ik} , resource growth rate, r_k , and mortality, m_i , were similar, while effects of temperature sensitivity in resource carrying capacity, K_k , and resource conversion efficiency, v_{ik} , differed (Figures S2C,D and S3C,D). Specifically, temperature-sensitive resource carrying capacity and conversion efficiency had qualitatively different effects on ND and FD within the 15°C temperature gradient when consumers' resource preferences were sufficiently different at the ambient temperature. That is, when

warming amplified the competitive dominance of one consumer species over the other (e.g., when N_1 had a stronger preference for its preferred resource than N_2 , and conversion efficiency of that preferred resource increased with warming), those consumers increased in ND over the 15°C temperature gradient, and the species' competitive trajectory pointed away from the neutral point (Figure S3D).

Across all analyses, temperature sensitivity of consumer mortality rate only affected FD. Competitive dynamics were highly insensitive to variation in the temperature sensitivity of mortality, as indicated by the absence of visible shifts in Figure 3E (see also Figure 4E).

3.3 | Question 2: How Do Thermal Asymmetries Contribute to Shifts in Competition With Warming?

We found that competing species that responded identically to warming (i.e., had identical temperature sensitivities for all parameters) experienced no change in competitive dynamics with warming (Figure S8). That is, without intra-process thermal asymmetries, inter-process thermal asymmetries did not affect competitive dynamics. However, when species differed in their temperature sensitivities for a given process, competitive dynamics shifted with warming. Within each parameter, larger differences in temperature sensitivities between competing species or between their resources (intra-process thermal asymmetries) were associated with larger shifts in competition, that is, larger Euclidean distances between the species' 'start' point at the ambient temperature and the 'end' point after 15°C warming (Figure 4, Figure S9). For consumer mortality rate, m_i , changes in position with warming were too small relative to that of other parameters to be visible in our analysis (Figure 4E).

For each parameter except consumption rate, warming created two distinct trajectories moving away from the ambient temperature point (Figure 4B–E). Which trajectory a species pair moved along depended on the relative values of the two focal temperature sensitivities, or the directionality of the intra-process thermal asymmetry (Figure 4B–E). For resource growth rate and consumer mortality rate, these trajectories moved monotonically toward the region of ND—FD space where a consumer's preferred resource was more abundant with warming, or where consumer mortality was minimized, though the latter effect was too small to observe for most species pairs. For example, if the temperature sensitivity for growth rate was higher for resource a than for resource b , then, when the system was warmed, the coexistence trajectory moved monotonically toward the region of competitive dominance for the species that preferentially consumed resource a , here N_2 (e.g., Figures 3B and 4B). Interestingly, the two trajectories for carrying capacity and conversion efficiency comprised one monotonic trajectory and one non-monotonic trajectory, both of which pointed ultimately toward neutrality (Figures 3C,D and 4C,D, Figure S3). However, large differences in ambient-temperature resource preferences between competing species drove qualitative changes to these trajectories over the 15°C warming gradient (Table S1, Figure S3).

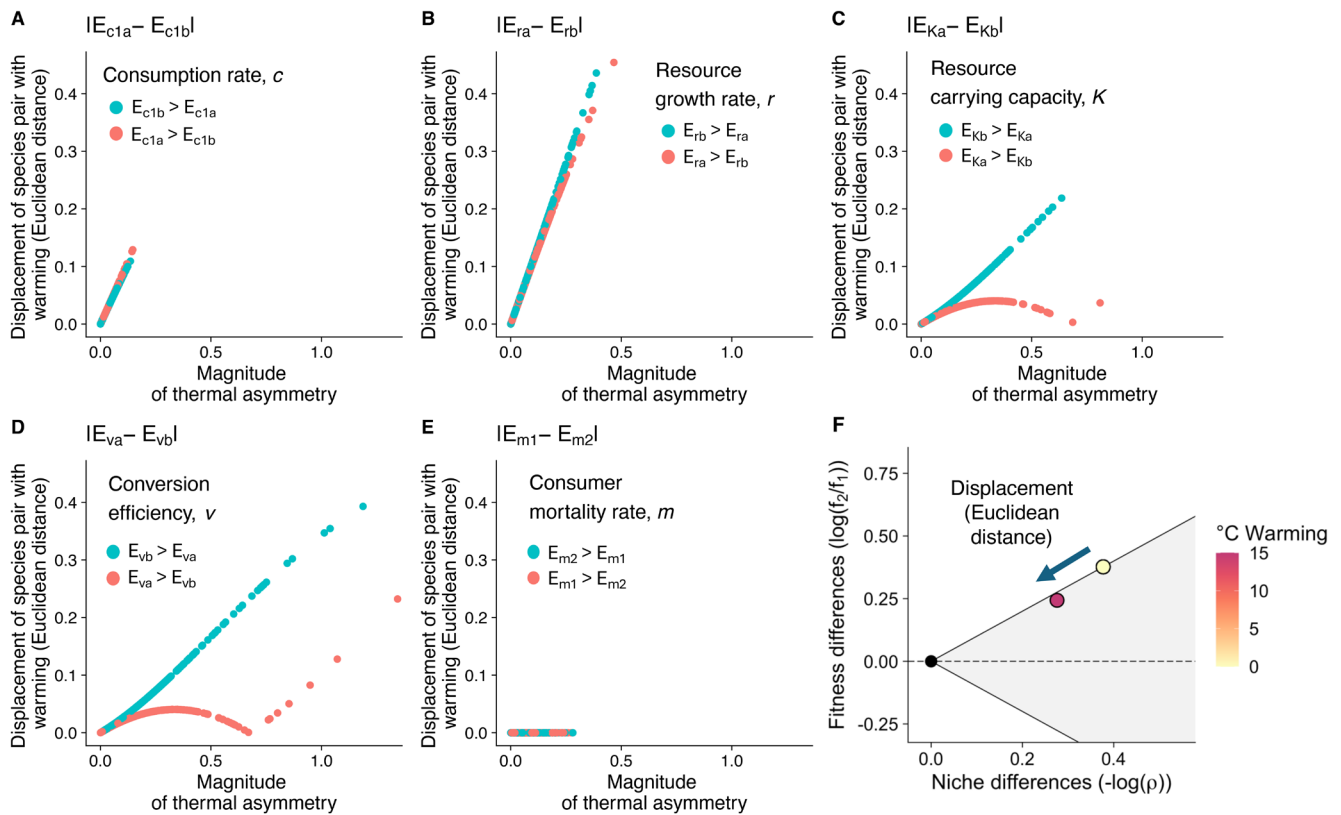


FIGURE 4 | Relationship between intra-process thermal asymmetries and shifts in competition with warming for each parameter. Panels A–E show displacement of species pairs with warming as a function of thermal asymmetries in each parameter in the consumer-resource model, and panel F depicts displacement of species pairs with warming on canonical niche difference-fitness difference axes. Thermal asymmetries are given as the absolute value of the difference between two draws of temperature sensitivities (one for each consumer or resource species) in each simulation and displacement is given as the Euclidean distance between a species pair's starting position and its position after 15°C warming. For each panel (A–E), only the focal temperature sensitivity parameter was allowed to vary; all other temperature sensitivity parameters were fixed at 0. Data points correspond to the difference in temperature sensitivities drawn for each of 500 simulations, and the colour of the points indicates the direction of the thermal asymmetry, that is, which E_v value is larger. For consumption rate (panel A), we vary only two of the four possible temperature sensitivities, here species N_1 's consumption rate on each resource, to display the effect of this thermal asymmetry most clearly.

3.4 | Question 3: How Does Warming Affect Competition via Effects on Niche and Fitness Differences?

For consumer species existing at the boundary of coexistence and competitive exclusion, the temperature sensitivity of all processes underlying competition generally drove decreases in both ND and FD with warming (Figure 5A, yellow dot to magenta dot). The majority of species pairs decreased in ND (58%) or FD (58%), and the median decreases in ND and FD were similar in magnitude (5.5% decrease and 9.7% decrease, respectively; Figure 5C,D). Declines in ND and FD with warming were associated with decreasing resource carrying capacity and resulting declines in all competition coefficients (Figure S10). This led to a general trend of competitive dynamics shifting toward neutrality with warming (Figure 5A). The distribution of shifts in competition with warming was skewed, and very large shifts were rare (Figure 5B). The magnitude of shift in competition was positively related to the magnitude of intra-process thermal asymmetries (Table S4). This result is reinforced by analysis of competition between species at the extremes of the observed temperature sensitivity distributions (top and bottom 5%), which

showed much larger shifts in competition with 15°C warming (Figure S11).

While some competitive dynamics became less neutral (i.e., increased in ND and FD) under 15°C warming, simulating extreme warming revealed that trajectories that initially moved away from neutrality bent back toward neutrality with sufficient warming (Figures S10 and S12). These results were robust to different consumer and resource traits and numbers of resource species, but the amount of warming required to shift the median effect toward neutrality under these different conditions varied (Figures S1–S4).

4 | Discussion

In this study, we drew from two prominent theories in ecology to address a central question in global change biology: how does warming affect competition for shared resources? Using empirically derived temperature sensitivities of the processes underlying resource competition, we found three general patterns in how simulated competition responded to warming. First, we found that warming tended to reduce both fitness differences

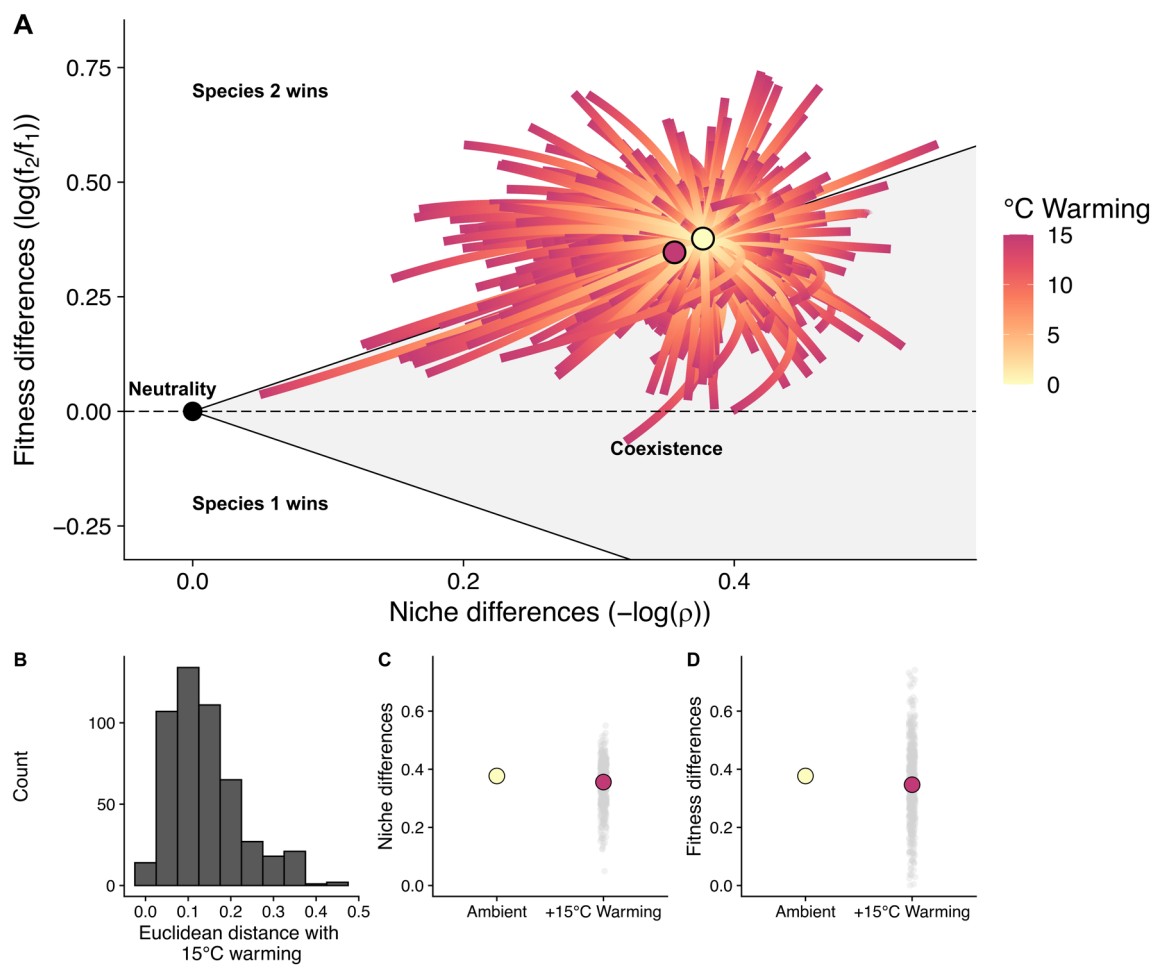


FIGURE 5 | Effects of warming on competition. Temperature sensitivities for all parameters are drawn from their distributions and then systems are warmed 15°C, and this simulation is repeated 500 times. (A) Simulation path for each draw of parameter values from the ambient temperature, 10°C (yellow dot), to 25°C (15°C warming). The median position along both axes after 15°C warming is plotted as a magenta dot. (B) Distribution of Euclidean distances between the position at the ambient temperature and after 15°C warming across 500 simulations. (C) Shifts in niche differences with warming. The median niche difference values (magenta dot) and the niche difference values for each simulation (grey dots) after 15°C warming are shown compared to the value at the ambient temperature (yellow dot). (D) Shifts in fitness differences with warming, displayed as in panel C.

and niche differences. This resulted in a general trend of warming causing species pairs to shift toward neutrality. Next, we found that warming-induced increases in mortality had surprisingly little effect on these shifts in competition with warming. Finally, we found that species pairs that underwent the largest shifts in competition with warming had highly asymmetric temperature sensitivities in the processes underlying competition. Together, these findings suggest that understanding how communities will change with warming requires an understanding of how both population vital rates and resource use traits change with temperature, and how variable these traits are within a community.

While much of the work seeking to understand biodiversity effects of climate change has focused on temperature-induced changes in fitness differences (e.g., growth rates; Ehrlén and Morris 2015; Kordas et al. 2011; Merow et al. 2014), here we show that understanding the effects of warming on competition requires an equal consideration of niche differences. As expected, warming-induced changes in resource growth rate had a large effect on competition in our analysis. This effect was near orthogonal to the coexistence boundary, occurring

primarily via shifts in fitness differences and causing resource growth rate to affect competitive hierarchies (by moving species from one region of ND—FD space to another) under warming. However, warming is unlikely to affect resource growth rate in isolation, and when warming was allowed to affect the full suite of processes underlying competition, species pairs generally also decreased in niche differences (Figure 5). These shifts were driven primarily by the effects of warming on consumption rate, resource carrying capacity, and consumer conversion efficiency—all of which determine patterns of relative resource use among species. Of these processes, warming effects on consumption rates also had the capacity to alter competitive hierarchies (Figure 3A). Together, these results add to a limited body of literature that suggests that temperature effects on niche differences play a key role in how warming affects communities (Adler et al. 2012).

Warming-induced changes to mortality rates have been a focal point of climate modelling and thermal physiology (Deutsch et al. 2008; Sunday et al. 2014), and have been shown to have effects on competition and, specifically, on fitness differences (Song et al. 2019). In our analysis, despite

exponentially increasing consumer mortality rates across the thermal gradient (Amarasekare and Savage 2012; Simon and Amarasekare 2024), effects of warming on mortality rate had almost no effect on competition. This is due to the independence of consumer mortality rate from effects of interspecific competition in consumer-resource models (Equation 6; Song et al. 2019), and is consistent with other work suggesting that mortality rates alone cannot drive shifts in coexistence (Chesson 2000; Smith and Amarasekare 2018). This suggests that, when species are living at temperatures far below their thermal optimum, and biological processes underlying competition are increasing exponentially with warming, quantifying the effects of warming on mortality may be less important than previously thought.

Within Modern Coexistence Theory, competition is considered neutral when there is an absence of both niche differences and fitness differences; biologically this means that species are functionally equivalent (Adler et al. 2007). In our analysis, warming had a general tendency to shift competing species toward neutrality, regardless of species' thermal traits under ambient temperatures (Figure 5, Figures S1–S5). In our simulations, this tendency to move toward neutrality as temperatures increased was related to the exponential decrease in resource carrying capacity with warming (Figure 2, Figure S10). As consumers became increasingly limited by reduced resource supply, intra- and inter-specific competition weakened, and consumers thus became more ecologically similar to one another. This result is consistent with theoretical (Hubbell 2001; Lande 1993) and empirical (Schreiber et al. 2022; Siepielski et al. 2010) work showing that smaller population sizes are associated with increased neutrality and stochasticity. Additionally, our finding that systems trend toward neutrality regardless of competitor traits is consistent with recent work suggesting that there may be characteristic trajectories for how competition changes under extrinsic influences, rather than bespoke shifts based on interspecific trait differences (Pastore et al. 2021).

While our finding that warming generally makes competition more neutral is independent of the model's starting conditions, we found that the specific resource-use and thermal traits of species influence the amount of warming required to shift competition toward neutrality (Figure 5, Figures S1–S4). That is, for some consumer pairs an increase in neutrality may not be realized within a biologically plausible range of temperatures, because the amount of warming required to create this effect might be large. In addition, if warming exceeds species' thermal optima for one or more of the processes underlying competition, then competitive trajectories may diverge from those predicted here (Figure S13). This is relevant in systems where species' thermal optima are close to mean environmental temperatures, and may also be relevant in systems with highly variable environmental temperatures, which at times exceed thermal optima (Amarasekare and Johnson 2017; Casas Goncalves and Amarasekare 2021; Smith and Amarasekare 2018).

Under changing thermal regimes, which types of species pairs are likely to experience large competitive shifts remains an open question. While most species pairs simulated in our study

experienced relatively small shifts in competitive dynamics with warming, species pairs that experienced large shifts were those that had large intra-process thermal asymmetries (Figure 4, Table S4, Figure S9). While a considerable body of work has described how inter-process thermal asymmetries can drive cross-scale responses to warming (e.g., Dell et al. 2014; Gibert et al. 2022; Gilbert et al. 2014; Lopez-Urrutia et al. 2006), in our analysis, these inter-process thermal asymmetries required accompanying intra-process thermal asymmetries in order to affect competition. For example, a higher temperature sensitivity of resource growth rate than conversion efficiency cannot, on its own, cause a shift in competition between species (Figure S8); there must also be interspecific (i.e., intra-process) variation in the temperature sensitivities of the growth rates or conversion efficiencies of the different resources to affect competition. Previous studies that have demonstrated generalized effects of warming on competition, have usually relied on differences in thermal optima among species to generate effects of warming on competition (Simon and Amarasekare 2024; Smith and Amarasekare 2018). Here, we show for the first time that general effects of warming on competition can emerge from interspecific differences in suboptimal thermal traits alone, demonstrating how warming changes competition even in conditions that are considered thermally benign for all species.

5 | Conclusions

Making general predictions for the effects of warming on communities has been a major challenge in ecology. The Metabolic Theory of Ecology, with its focus on the temperature sensitivity of metabolic processes across biological scales, and Modern Coexistence Theory, through its conceptual framework for species competition, have both advanced our understanding of ecosystem responses to environmental change. However, differences in the domains of these two fields, one of which focuses on interspecific trait differences and the other on macroecological differences in key metabolic processes as drivers of change, have prevented their integration to date. Here we integrated concepts from both fields and combined this with empirical thermal trait data to develop new, general theory for how competition changes with warming and to advance a more coherent understanding of global change. The next step will be to test these predictions empirically. Doing so will facilitate better linkages between general patterns in biological responses to warming and context-specific stewardship strategies under rapid environmental change.

Author Contributions

K.E.D., J.R.B., T.N.G., P.-J.K., P.L.T. and M.I.O. conceptualized the study. J.R.B. curated the dataset. K.E.D., J.R.B. and P.-J.K. developed software and K.E.D. analysed the data. K.E.D. wrote the original manuscript. All authors contributed to reviewing and editing the manuscript. J.R.B. supervised the study.

Acknowledgements

J.R.B., T.N.G. and M.I.O. were supported by Discovery Grants from the Natural Science and Engineering Research Council of Canada. P.-J.K.

was supported by the Yushan Fellow Program, Ministry of Education, Taiwan (MOE-110-YSFAG-0003-001-P1). K.E.D. was supported by a Centre for Ecosystem Management Postdoctoral Fellowship. We thank Megan Szojka, Jurg W. Spaak, and two anonymous reviewers for helpful feedback on the manuscript.

Funding

This work was supported by Natural Sciences and Engineering Research Council of Canada. Centre for Ecosystem Management, University of Guelph.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at the DOI <https://zenodo.org/records/19634310>.

Peer Review

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/ele.70395>.

References

- Adler, P. B., H. J. Dalglish, and S. P. Ellner. 2012. "Forecasting Plant Community Impacts of Climate Variability and Change: When Do Competitive Interactions Matter?" *Journal of Ecology* 100: 478–487.
- Adler, P. B., J. HilleRisLambers, and J. M. Levine. 2007. "A Niche for Neutrality." *Ecology Letters* 10: 95–104.
- Alexander, J. M., J. M. Diez, and J. M. Levine. 2015. "Novel Competitors Shape Species' Responses to Climate Change." *Nature* 525: 515–518.
- Amarasekare, P., and C. Johnson. 2017. "Evolution of Thermal Reaction Norms in Seasonally Varying Environments." *American Naturalist* 189: E31–E45.
- Amarasekare, P., and V. Savage. 2012. "A Framework for Elucidating the Temperature Dependence of Fitness." *American Naturalist* 179: 178–191.
- Bernhardt, J. R., J. M. Sunday, and M. I. O'Connor. 2018. "Metabolic Theory and the Temperature Size Rule Explain the Temperature Dependence of Population Carrying Capacity." *American Naturalist* 192: 187–197.
- Bernhardt, J. R., J. M. Sunday, P. L. Thompson, and M. I. O'Connor. 2018. "Nonlinear Averaging of Thermal Experience Predicts Population Growth Rates in a Thermally Variable Environment." *Proceedings of the Royal Society B: Biological Sciences* 285: 20181076.
- Brown, J. H., J. F. Gillooly, A. P. Allen, V. M. Savage, and G. B. West. 2004. "Toward a Metabolic Theory of Ecology." *Ecology* 85: 1771–1789.
- Casas Goncalves, G., and P. Amarasekare. 2021. "Persistence of Tri-Trophic Interactions in Seasonal Environments." *Journal of Animal Ecology* 90: 298–310.
- Chesson, P. 1990. "MacArthur's Consumer-Resource Model." *Theoretical Population Biology* 37: 26–38.
- Chesson, P. 2000. "General Theory of Competitive Coexistence in Spatially-Varying Environments." *Theoretical Population Biology* 58: 211–237.
- Chesson, P. 2011. "Ecological Niches and Diversity Maintenance." In *Biodiversity: Models and Applications*, 43–60. InTech.
- Chesson, P., and J. J. Kuang. 2008. "The Interaction Between Predation and Competition." *Nature* 456: 235–238.
- Clegg, T., and S. Pawar. 2024. "Variation in Thermal Physiology Can Drive the Temperature-Dependence of Microbial Community Richness." *eLife* 13: e84662.
- Coblentz, K. E., M. Novak, and J. P. DeLong. 2023. "Predator Feeding Rates May Often Be Unsaturated Under Typical Prey Densities." *Ecology Letters* 26: 302–312.
- Dell, A. I., S. Pawar, and V. M. Savage. 2011. "Systematic Variation in the Temperature Dependence of Physiological and Ecological Traits." *Proceedings of the National Academy of Sciences* 108: 10591–10596.
- Dell, A. I., S. Pawar, and V. M. Savage. 2014. "Temperature Dependence of Trophic Interactions Are Driven by Asymmetry of Species Responses and Foraging Strategy." *Journal of Animal Ecology* 83: 70–84.
- Deutsch, C. A., J. J. Tewksbury, R. B. Huey, et al. 2008. "Impacts of Climate Warming on Terrestrial Ectotherms Across Latitude." *Proceedings of the National Academy of Sciences* 105: 6668–6672.
- Ehrlén, J., and W. F. Morris. 2015. "Predicting Changes in the Distribution and Abundance of Species Under Environmental Change." *Ecology Letters* 18: 303–314.
- Enquist, B. J., E. P. Economo, T. E. Huxman, A. P. Allen, D. D. Ignace, and J. F. Gillooly. 2003. "Scaling Metabolism From Organisms to Ecosystems." *Nature* 423: 639–642.
- Germain, R. M., M. M. Mayfield, and B. Gilbert. 2018. "The 'Filtering' Metaphor Revisited: Competition and Environment Jointly Structure Invasibility and Coexistence." *Biology Letters* 14: 20180460.
- Gibert, J. P., J. M. Grady, and A. I. Dell. 2022. "Food Web Consequences of Thermal Asymmetries." *Functional Ecology* 36: 1887–1899.
- Gilbert, B., T. D. Tunney, K. S. McCann, et al. 2014. "A Bioenergetic Framework for the Temperature Dependence of Trophic Interactions." *Ecology Letters* 17: 902–914.
- Godoy, O. 2019. "Coexistence Theory as a Tool to Understand Biological Invasions in Species Interaction Networks: Implications for the Study of Novel Ecosystems." *Functional Ecology* 33: 1190–1201.
- Godoy, O., and J. M. Levine. 2014. "Phenology Effects on Invasion Success: Insights From Coupling Field Experiments to Coexistence Theory." *Ecology* 95: 726–736.
- Godwin, C. M., F.-H. Chang, and B. J. Cardinale. 2020. "An Empiricist's Guide to Modern Coexistence Theory for Competitive Communities." *Oikos* 129: 1109–1127.
- Grainger, T. N., J. M. Levine, and B. Gilbert. 2019. "The Invasion Criterion: A Common Currency for Ecological Research." *Trends in Ecology & Evolution* 34: 925–935.
- Hubbell, S. P. 2001. "The Unified Neutral Theory of Biodiversity and Biogeography." In *The Unified Neutral Theory of Biodiversity and Biogeography*. Princeton University Press.
- Johnson, C. A., P. Dutt, and J. M. Levine. 2022. "Competition for Pollinators Destabilizes Plant Coexistence." *Nature* 607: 721–725.
- Kingsolver, J. G. 2009. "The Well-Tempered Biologist: (American Society of Naturalists Presidential Address)." *American Naturalist* 174: 755–768.
- Knies, J. L., and J. G. Kingsolver. 2010. "Erroneous Arrhenius: Modified Arrhenius Model Best Explains the Temperature Dependence of Ectotherm Fitness." *American Naturalist* 176: 227–233.
- Kordas, R. L., C. D. G. G. Harley, and M. I. O'Connor. 2011. "Community Ecology in a Warming World: The Influence of Temperature on Interspecific Interactions in Marine Systems." *Journal of Experimental Marine Biology and Ecology* 400: 218–226.
- Kraft, N. J. B., R. Valencia, and D. D. Ackerly. 2008. "Functional Traits and Niche-Based Tree Community Assembly in an Amazonian Forest." *Science* 322: 580–582.
- Lande, R. 1993. "Risks of Population Extinction From Demographic and Environmental Stochasticity and Random Catastrophes." *American Naturalist* 142: 911–927.

- Letten, A. D., P. J. Ke, and T. Fukami. 2017. "Linking Modern Coexistence Theory and Contemporary Niche Theory." *Ecological Monographs* 87: 161–177.
- Lopez-Urrutia, A., E. San Martin, R. P. Harris, and X. Irigoien. 2006. "Scaling the Metabolic Balance of the Oceans." *Proceedings of the National Academy of Sciences* 103: 8739–8744.
- MacArthur, R. 1970. "Species Packing and Competitive Equilibrium for Many Species." *Theoretical Population Biology* 1: 1–11.
- Magee, L. J., D. J. B. Smith, D. Bauman, et al. 2025. "Memories of Trees Past: Coexistence Implications of Legacy Conspecific Density Dependence." *Ecology Letters* 28: e70197.
- Martin, T. L., and R. B. Huey. 2008. "Why 'Suboptimal' Is Optimal: Jensen's Inequality and Ectotherm Thermal Preferences." *American Naturalist* 171: E102–E118.
- Merow, C., A. M. Latimer, A. M. Wilson, S. M. McMahon, A. G. Rebelo, and J. A. Silander Jr. 2014. "On Using Integral Projection Models to Generate Demographically Driven Predictions of Species' Distributions: Development and Validation Using Sparse Data." *Ecography* 37: 1167–1183.
- Michaletz, S. T., and J. C. Garen. 2024. "Hotter Is Not (Always) Better: Embracing Unimodal Scaling of Biological Rates With Temperature." *Ecology Letters* 27: e14381.
- Novak, M., K. E. Coblenz, and J. P. DeLong. 2025. "In Defense of Type I Functional Responses: The Frequency and Population Dynamic Effects of Feeding on Multiple Prey at a Time." *American Naturalist* 206: 347–361.
- O'Connor, M. I. 2009. "Warming Strengthens an Herbivore: Plant Interaction." *Ecology* 90: 388–398.
- O'Connor, M. I., B. Gilbert, and C. Brown. 2011. "Theoretical Predictions for How Temperature Affects the Dynamics of Interacting Herbivores and Plants." *American Naturalist* 178: 626–638.
- Parmesan, C., and G. Yohe. 2003. "A Globally Coherent Fingerprint of Climate Change Impacts Across Natural Systems." *Nature* 421: 37–42.
- Pastore, A. I., G. Barabás, M. D. Bimler, M. M. Mayfield, and T. E. Miller. 2021. "The Evolution of Niche Overlap and Competitive Differences." *Nature Ecology & Evolution* 5: 330–337.
- Pawar, S., A. I. Dell, V. M. Savage, and J. L. Knies. 2016. "Real Versus Artificial Variation in the Thermal Sensitivity of Biological Traits." *American Naturalist* 187: E41–E52.
- Pérez-Ramos, I. M., L. Matías, L. Gómez-Aparicio, and Ó. Godoy. 2019. "Functional Traits and Phenotypic Plasticity Modulate Species Coexistence Across Contrasting Climatic Conditions." *Nature Communications* 10: 2555.
- Peters, R. L., and T. E. Lovejoy. 1992. *Global Warming and Biological Diversity*. Yale University Press.
- Rall, B. C., O. Vucic-Pestic, R. B. Ehnes, M. Emmerson, and U. Brose. 2010. "Temperature, Predator-Prey Interaction Strength and Population Stability: Temperature and Interaction Strength." *Global Change Biology* 16: 2145–2157.
- Román-Palacios, C., and J. J. Wiens. 2020. "Recent Responses to Climate Change Reveal the Drivers of Species Extinction and Survival." *Proceedings of the National Academy of Sciences of the United States of America* 117: 4211–4217.
- Savage, V. M., J. F. Gillooly, J. H. Brown, G. B. West, and E. L. Charnov. 2004. "Effects of Body Size and Temperature on Population Growth." *American Naturalist* 13: 429–441.
- Schreiber, S. J., J. M. Levine, O. Godoy, N. J. B. Kraft, and S. P. Hart. 2022. "Does Deterministic Coexistence Theory Matter in a Finite World?" *Ecology* 104: e3838.
- Siepielski, A. M., K.-L. Hung, E. E. B. Bein, and M. A. McPeck. 2010. "Experimental Evidence for Neutral Community Dynamics Governing an Insect Assemblage." *Ecology* 91: 847–857.
- Simon, M. W., and P. Amarasekare. 2024. "Predicting the Fundamental Thermal Niche of Ectotherms." *Ecology* 105: e4289.
- Smith, D. J., and P. Amarasekare. 2018. "Toward a Mechanistic Understanding of Thermal Niche Partitioning." *American Naturalist* 191: E57–E75.
- Song, C., G. Barabás, and S. Saavedra. 2019. "On the Consequences of the Interdependence of Stabilizing and Equalizing Mechanisms." *American Naturalist* 194: 627–639.
- Spaak, J. W., and F. De Laender. 2020. "Intuitive and Broadly Applicable Definitions of Niche and Fitness Differences." *Ecology Letters* 23: 1117–1128.
- Spaak, J. W., P.-J. Ke, A. D. Letten, and F. De Laender. 2023. "Different Measures of Niche and Fitness Differences Tell Different Tales." *Oikos* 2023: e09573.
- Sunday, J. M., A. E. Bates, M. R. Kearney, et al. 2014. "Thermal-Safety Margins and the Necessity of Thermoregulatory Behavior Across Latitude and Elevation." *Proceedings of the National Academy of Sciences* 111: 5610–5615.
- Terry, J. C. D. 2025. "An Experimental Validation Test of Ecological Coexistence Theory to Forecast Extinction Under Rising Temperatures." *Ecology Letters* 28: e70047.
- Tilman, D. 1994. "Competition and Biodiversity in Spatially Structured Habitats." *Ecology* 75: 2–16.
- Usinowicz, J., and J. M. Levine. 2018. "Species Persistence Under Climate Change: A Geographical Scale Coexistence Problem." *Ecology Letters* 21: 1589–1603.
- Vasseur, D. A., and K. S. McCann. 2005. "A Mechanistic Approach for Modeling Temperature-Dependent Consumer-Resource Dynamics." *American Naturalist* 166: 184–198.
- Wan, J., P.-J. Ke, I. Hordijk, L. Bialic-Murphy, and T. W. Crowther. 2025. "Functional Coexistence Theory: Identifying Mechanisms Linking Biodiversity and Ecosystem Function." *Ecological Monographs* 95: e70033.
- Weiher, E., G. D. P. Clarke, and P. A. Keddy. 1998. "Community Assembly Rules, Morphological Dispersion, and the Coexistence of Plant Species." *Oikos* 81: 309–322.
- Welter, J. R., J. P. Benstead, W. F. Cross, et al. 2015. "Does N₂ Fixation Amplify the Temperature Dependence of Ecosystem Metabolism?" *Ecology* 92: 603–610.
- Yamamichi, M., T. Gibbs, and J. M. Levine. 2022. "Integrating Eco-Evolutionary Dynamics and Modern Coexistence Theory." *Ecology Letters* 25: 2091–2106.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Starting conditions for each competition scenario. Each scenario is implemented in the model (Equation 4) through a set of values at the ambient temperature, that is, $Y_{T_{amb}}$ values in the temperature-sensitivity function (Equation 9), for example, $c_{1a} = c_{1a_{T_{amb}}}$. **Table S2:** Demographics of the dataset in terms of cellularity and trophy. **Table S3:** Observed ranges compared to 95% credible intervals for the temperature sensitivities of each parameter. **Table S4:** Partial regression results for relationships between thermal asymmetries in consumer or resource parameters and displacement of species pairs in the main text simulation (Figure 5). Displacement is measured as the Euclidean distance between the species position in NFD space at 10°C (reference temperature) and at 25°C. Significant relationships ($p < 0.05$)

are given in bold. **Figure S1:** Effects of warming on competition in 2-consumer, 8-resource systems. **Figure S2:** Simulations of warming in species pairs with even resource preferences, where each resource has the same growth rate at ambient temperature (Table S1, Scenario 2). **Figure S3:** Simulations of warming in species pairs with highly uneven resource preferences, where each resource has the same growth rate at ambient temperature (Table S1, Scenario 3). **Figure S4:** Effects of warming on competition for species pairs with asymmetric resource competition. **Figure S5:** Effects of 5°C warming on competition. **Figure S6:** Different random draw structures for the temperature sensitivity of consumption rate when all non-focal processes are not temperature sensitive. **Figure S7:** Systematic variation in temperature sensitivities with cellularity. **Figure S8:** Effects of warming on competition when inter-process thermal asymmetries are present, but intra-process thermal asymmetries are not. **Figure S9:** Effects of thermal asymmetries in each parameter (raw values) on Euclidean (Eucl) distance, fitness differences (FD) and niche differences (ND). **Figure S10:** Median trajectories of key model terms with 200°C warming. **Figure S11:** Effects of extreme thermal asymmetries on competition under warming. **Figure S12:** Effects of extreme warming (50°C) on competition. **Figure S13:** Effects of unimodal temperature sensitivity on study results.